Cystic fibrosis and ER quality control

- **CFTR (Cystic Fibrosis Transmembrane Conductance Regulator)**
- ATP-binding cassette (ABC) transporter
- Functions as a **Cl^- channel** in epithelial cells

- **Physiological function**
- Ion and water transport
- Maintains mucus hydration
→ Thin, hydrated mucus 유지

- **ER Folding & Quality Control**
- CFTR is synthesized and folded in the ER
- Undergoes N-linked glycosylation
- Interacts with chaperones (BiP, calnexin)

- **Genetic disorder caused by CFTR mutation**
- Deletion of phenylalanine at position 508 (most common)
- Causes **misfolding in the ER** → ER에서 QC에 의해 제거됨
- No functional CFTR at membrane ↓ Cl^- secretion
  → ↓ water movement → Thick, dehydrated mucus (lung, pancreas, intestine)
- Cystic fibrosis는 유럽계 인구에서 높은 빈도로 나타나는 유전 질환 (약 2,500–3,500명 중 1명)

CHAPTER CONTENTS

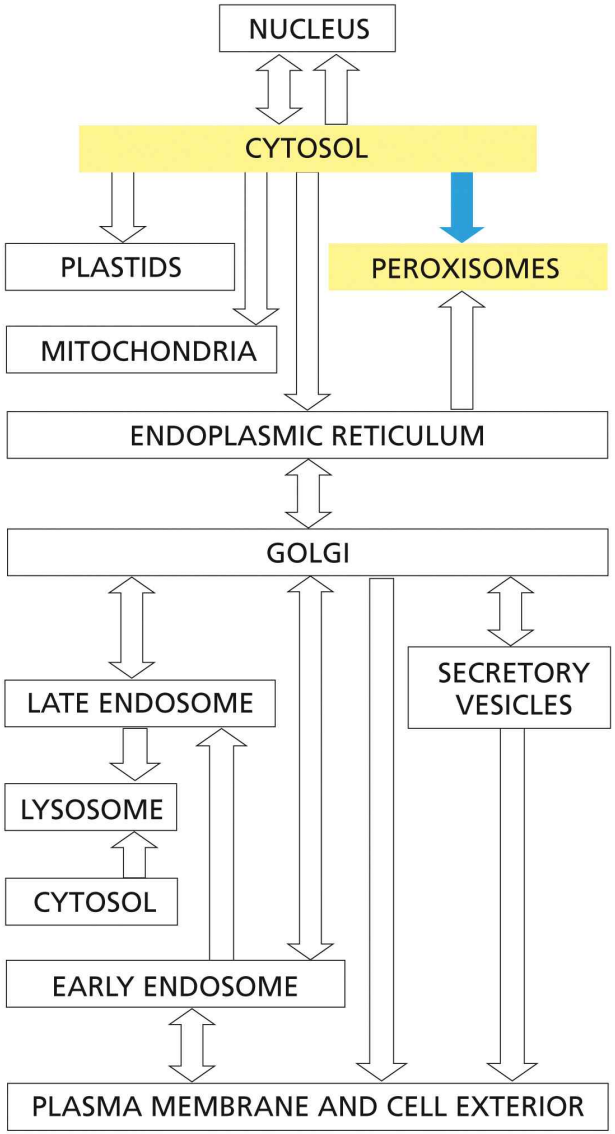
THE COMPARTMENTALIZATION OF CELLS

THE ENDOPLASMIC RETICULUM

PEROXISOMES

THE TRANSPORT OF PROTEINS INTO MITOCHONDRIA AND CHLOROPLASTS

THE TRANSPORT OF MOLECULES BETWEEN THE NUCLEUS AND THE CYTOSOL


NUCLEUS
CYTOSOL
PLASTIDS
PEROXISOMES
MITOCHONDRIA
ENDOPLASMIC RETICULUM
GOLGI
LATE ENDOSOME
SECRETORY VESICLES
LYSOSOME
CYTOSOL
EARLY ENDOSOME
PLASMA MEMBRANE AND CELL EXTERIOR

Peroxisomes Use Molecular Oxygen and Hydrogen Peroxide to Perform Oxidation Reactions

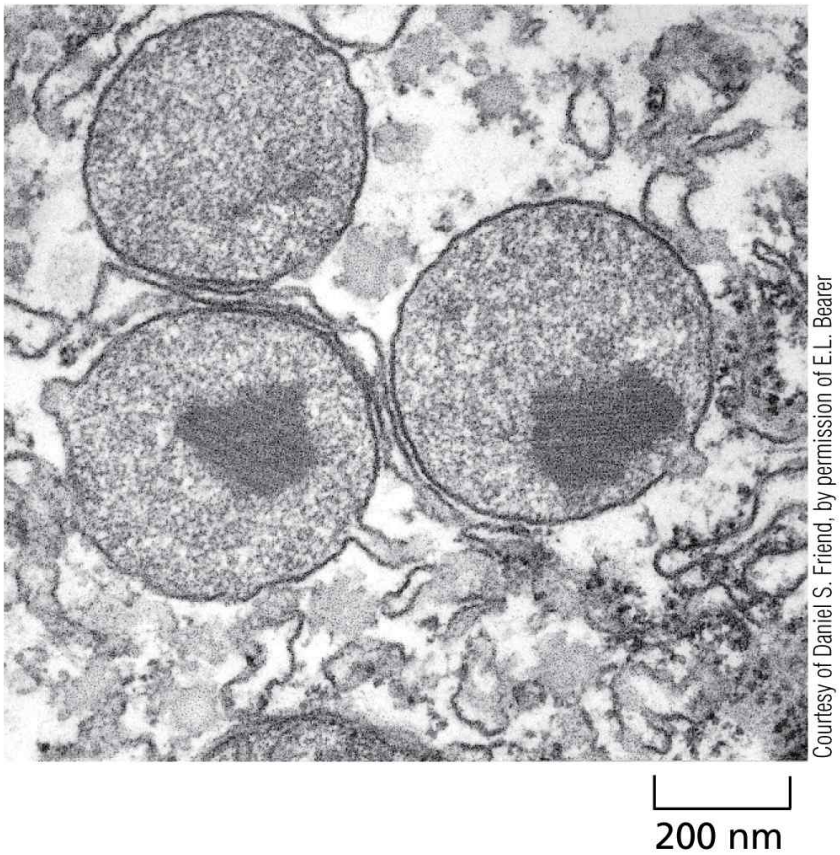


Figure 12–43 An electron micrograph of three peroxisomes in a rat liver cell.
The paracrystalline, electron-dense inclusions are composed primarily of the enzyme urate oxidase.

- **Single-membrane organelles**
  DNA, ribosome 없음 → 모든 단백질은 핵에서 유래
- 전자현미경에서 **catalase, oxidase** 축적 → Crystalloid core 관찰 가능
- **Oxidation**
  - $RH_2 + O_2 \rightarrow R + H_2O_2$ by **oxidase**
  - $2H_2O_2 \rightarrow 2H_2O + O_2$ by **catalase** (H_2O_2 제거 및 peroxidation 반응 수행)
- **Detoxification (간에서 중요)**
  - 알코올, 포름산, 독성 물질 분해
  - 섭취한 ethanol의 약 25% peroxisome에서 acetaldehyde (CH_3CHO)로 산화
- **Fatty acid b-oxidation**
  - 지방산 → acetyl-CoA → cytosol export
  - 동물 : mitochondria + peroxisome
  - 효모 식물 peroxisome only
- **Protein transport**
  - 단백질은 cytosol 합성 후 folded 상태로 import 가능
  - 일부 ER을 거쳐 peroxisome 막으로 이동
  - PEX proteins (peroxins) receptor에 의해 peroxisome으로 운반

Plasmalogen

- Phospholipid의 한 종류로, 세포막 특히 신경세포의 myelin sheath(수초), 심장, 면역세포에 풍부
- Myelin sheath 인지질의 약 80–90%를 차지
- 막의 안정성, 항산화 기능, 신호전달에 중요한 역할

- Peroxisome의 역할
- Plasmalogen 합성 경로의 초기단계가 퍼옥시좀에서 일어남
- 이후 중간체는 ER 소포체로 이동하여 합성이 완성됨.

- 신경세포와 마이엘린 수초
- Plasmalogen은 특히 신경세포 마이엘린 수초(myelin sheath)의 주요 구성 성분.
- 따라서 peroxisome 결함은 신경전달 속도 저하, 신경퇴행성 질환 발생

- Peroxisome 질환 (예: Zellweger syndrome, Rhizomelic chondrodysplasia punctata)
- Plasmalogen 합성이 제대로 안 되어 신경, 근육, 호흡기, 심장 등의 기능 이상

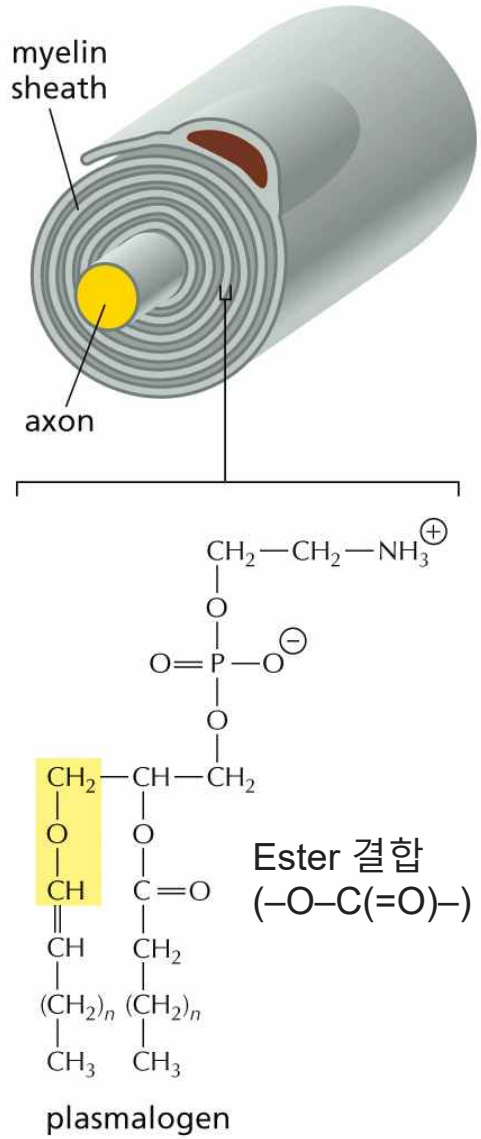


Figure 12–44 The structure of a plasmalogen

Zellweger syndrome Peroxisome disease

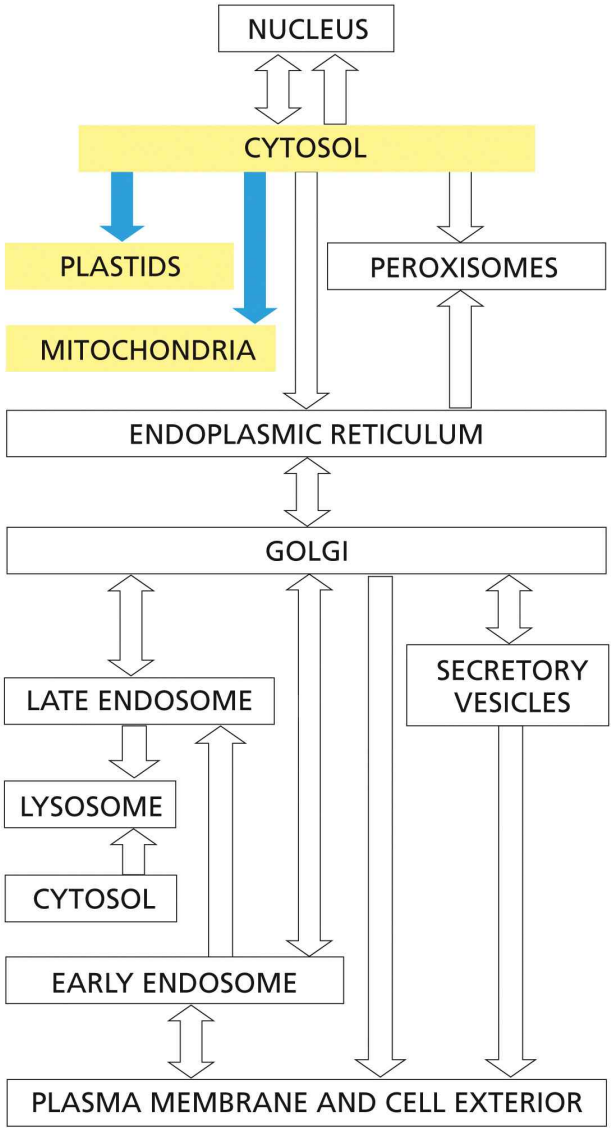
- PEX genes
- Peroxisome 형성과 기능에 필수적인 단백질 (peroxin)
- Mutation 발생 시 → Peroxisome이 제대로 만들어지지 않거나 기능 상실

- Cellular Changes
- Peroxisome 없음 또는 비정상
- Matrix protein import 실패
- lipid metabolism 장애

- Metabolic Consequences
- Peroxisomal β-oxidation 감소 → Very long-chain fatty acids (VLCFA) 축적
- plasmalogen 감소 → Membrane structure/function 이상
- H_2O_2 detox 문제 → Oxidative stress 증가

- Clinical Features
- Severe hypotonia
- Seizures
- Hepatomegaly / liver dysfunction
- Craniofacial dysmorphism (high forehead, flat face)

→ 대부분 infancy에 사망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PLASMA MEMBRANE AND CELL EXTERIOR

- **Mitochondria**
- **Double-membrane structure (outer + inner membrane)**
- Cristae는 inner membrane의 일부 → increased surface area for oxidative phosphorylation
- Matrix contains TCA cycle enzymes, mitochondrial DNA, and ribosomes

→ **ATP production via oxidative phosphorylation**

- **Chloroplasts**
- Double-membrane organelles
- Internal **thylakoid membrane system** forms grana stacks
- Stroma contains Calvin cycle enzymes, chloroplast DNA, and ribosomes
- Photosynthesis occurs in **thylakoid membranes** (light reactions)
- calvin cycle → stroma

→ **ATP + NADPH production via photosynthesis**

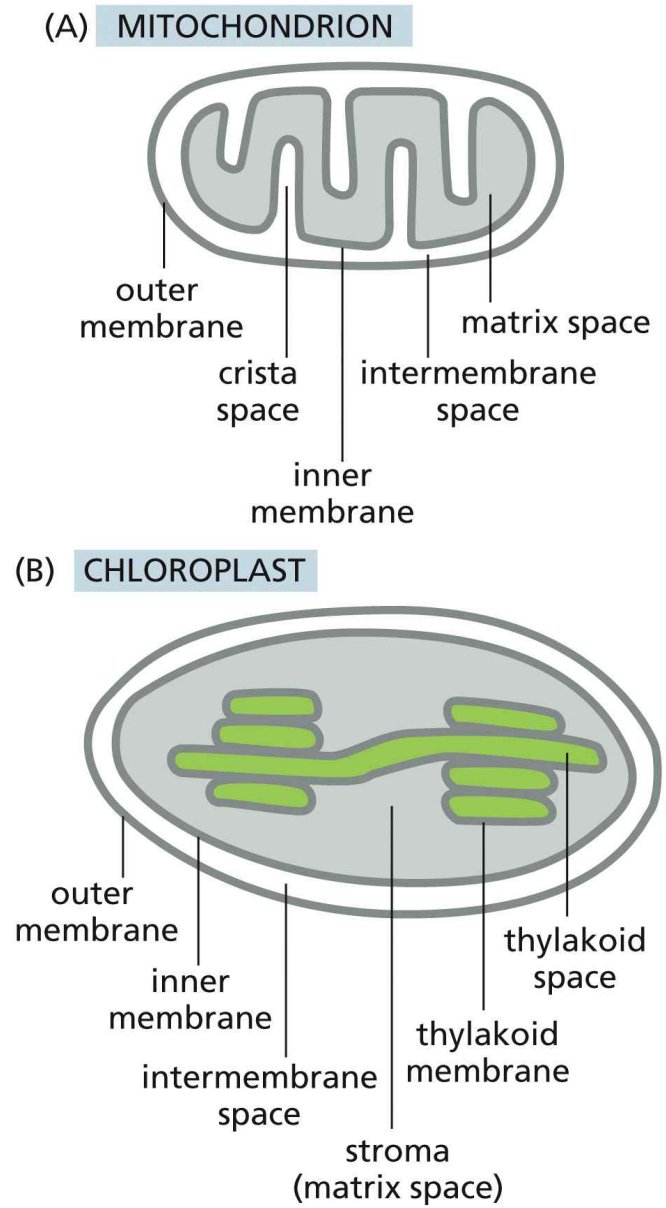


Figure 12–47 The subcompartments of mitochondria and chloroplasts. In contrast to the cristae of mitochondria (**A**), the thylakoids of chloroplasts (**B**) are not connected to the inner membrane and therefore form a sealed continuous compartment with a separate internal space.

Translocation Into Mitochondria Depends on Signal Sequences and Protein Translocators

- **대부분의 mitochondrial protein은 cytosol에서 합성 → translocator complex 통해 import**
- 반드시 단백질 **unfolded 상태 필요**

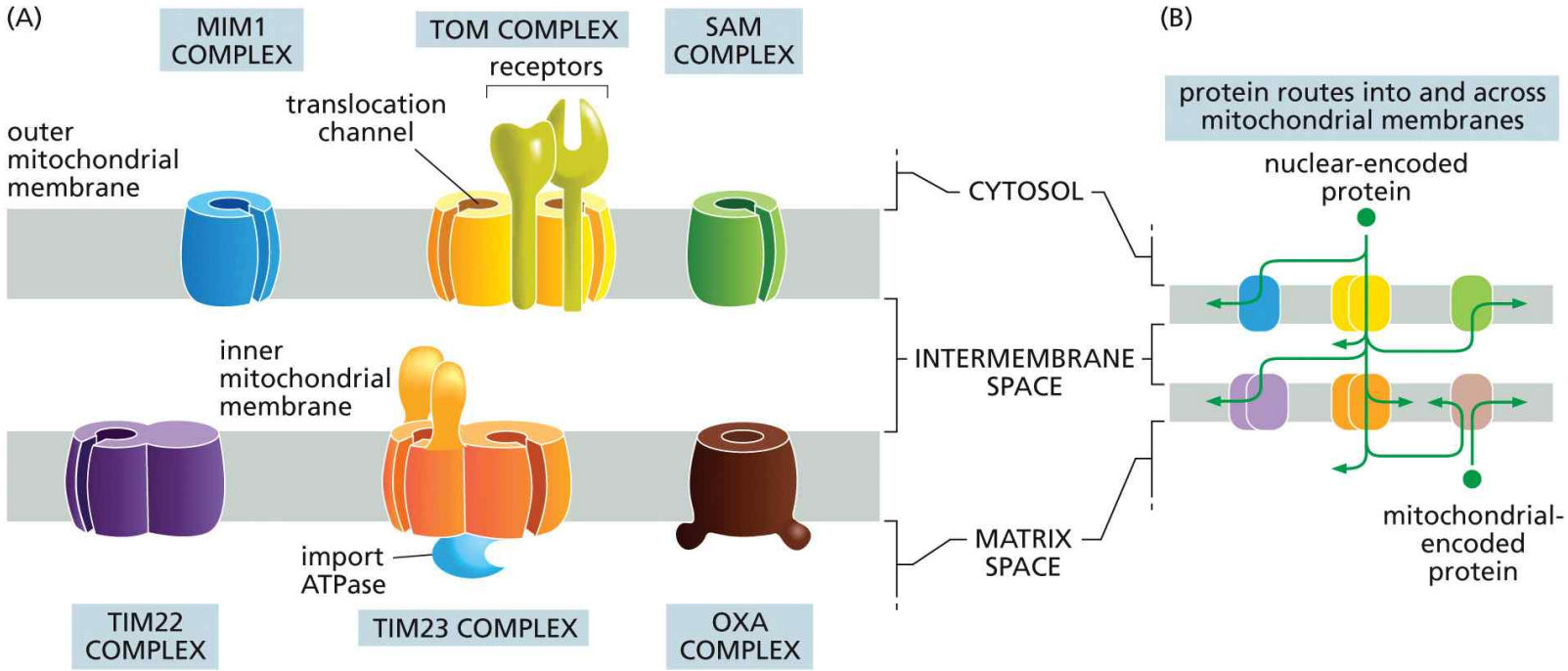
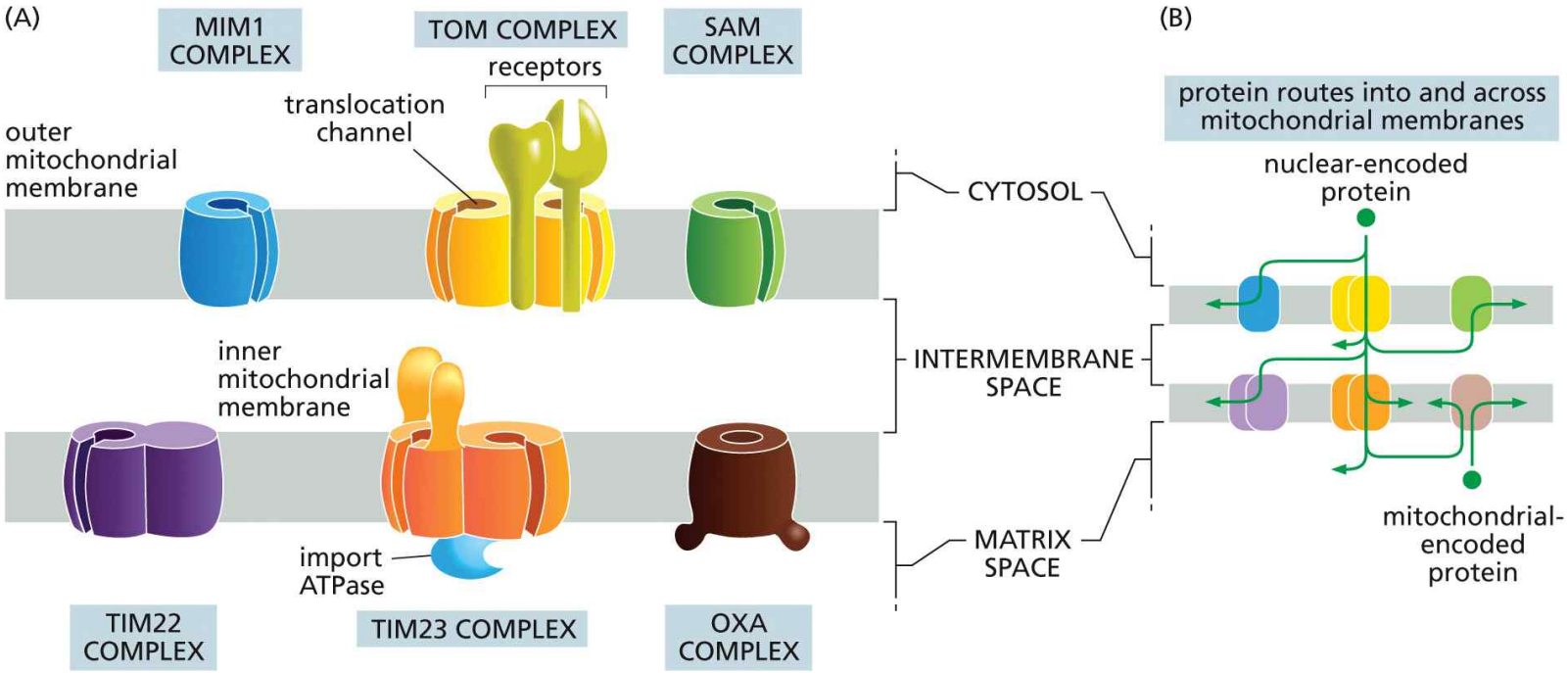


Figure 12–48 The protein translocators in the mitochondrial membranes.

- **TOM complex** (Translocase of Outer Membrane): Entry gate
- Cytosolic protein 인식 → 미토콘드리아 outer membrane 통과
- **TIM23 complex**: Inner membrane translocation into matrix 과 일부 inner membrane protein
- **mtHsp70 (ATP 사용)** → 단백질을 당겨서 matrix로 이동 (반드시 unfolded 상태 필요)

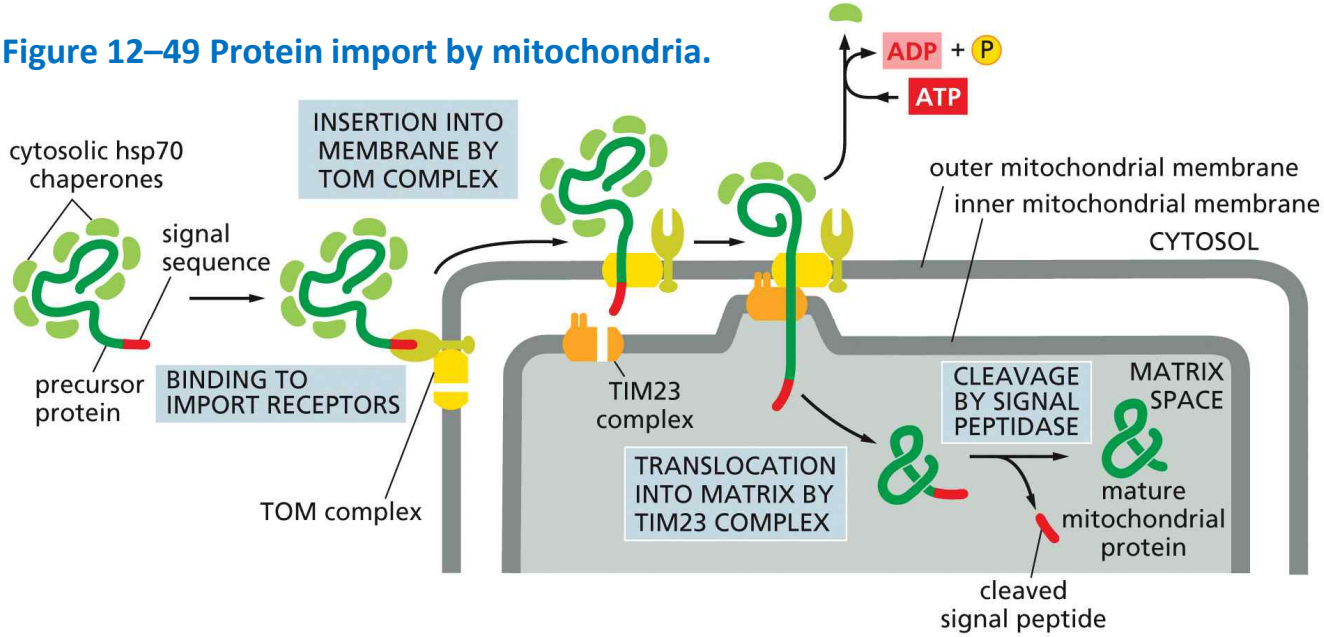
Translocation Into Mitochondria Depends on Signal Sequences and Protein Translocators

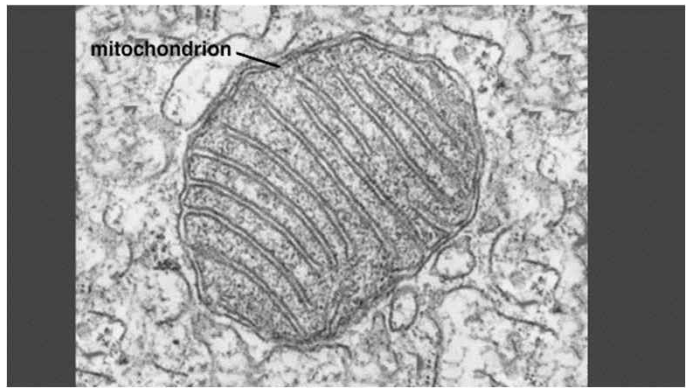


- **TIM22 complex**: Inserts multi-pass proteins into inner membrane
- **SAM complex**: Folding and insertion of β-barrel proteins in outer membrane (TOM → SAM 경로)
- **MIM complex**: Inserts α-helical proteins into outer membrane
- **OXA complex**: Inserts mitochondrially encoded proteins into inner membrane
  (matrix에서 합성된 단백질 → inner membrane insertion)

Mitochondrial Proteins Are Imported Post-translationally as Unfolded Polypeptide Chains

Figure 12–49 Protein import by mitochondria.



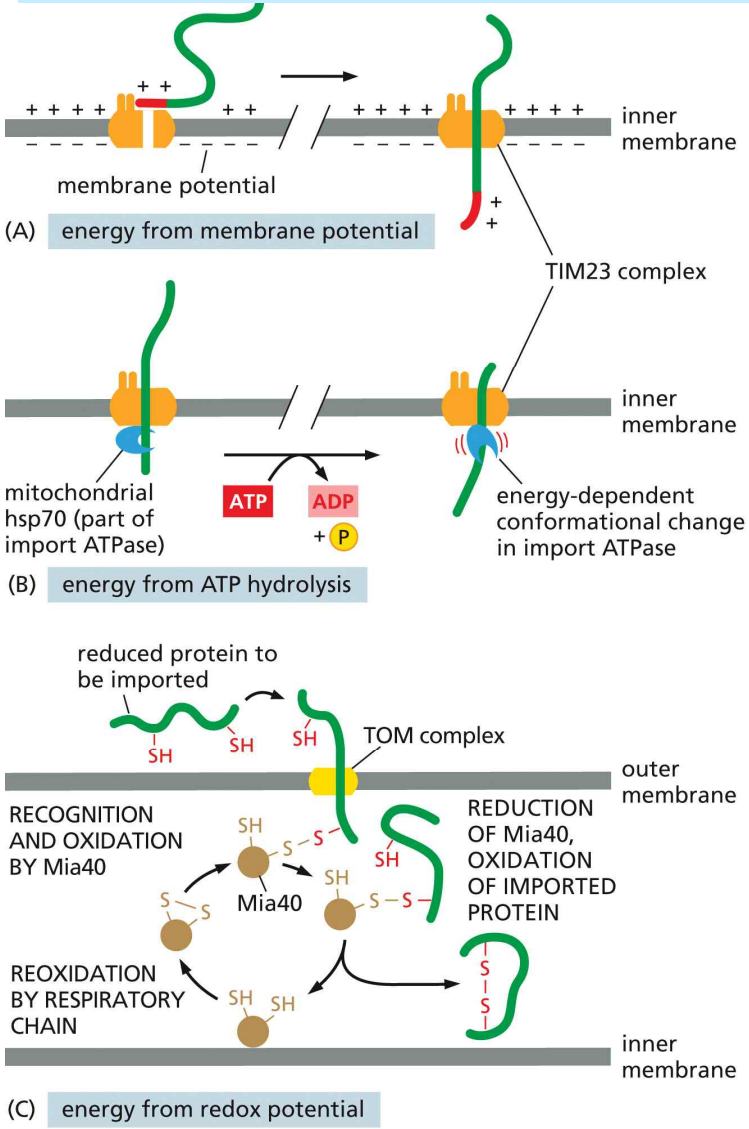


- **Key Steps**

1. Precursor proteins are synthesized in cytosol with an **N-terminal signal sequence**
2. Maintained in an **unfolded state** and bound by cytosolic **Hsp70 chaperones**
3. Signal sequence is recognized by the **TOM complex** (outer membrane)
4. Protein is translocated across both membranes via the **TIM23 complex** (inner membrane)
5. Protein is pulled into the matrix by **mitochondrial Hsp70** (ATP-dependent)
6. The Signal peptide is cleaved by a **signal peptidase** in the matrix
7. The Protein folds into **functional mitochondrial protein** (maturation)

➢ **Energy-dependent process**: Requires ATP hydrolysis by hsp70

Protein Import Is Powered by ATP Hydrolysis, a Membrane Potential, and Redox Potential



- **Membrane potential (Δψ)**
- TIM23 복합체를 통한 signal sequence 당김
- **Positively charged N-terminal signal sequence**가 막전위 의존적으로 matrix 방향으로 이동

- **ATP hydrolysis (mtHsp70 chaperone)**
- 미토콘드리아 matrix의 hsp70 (import ATPase complex)
- ATP hydrolysis 사용→ polypeptide를 순차적으로 당김

→ **Unfolding** 유지 및 matrix 방향으로 directionality 제공

- **Redox potential (Mia40 system)**
- Intermembrane space에서 cysteine-containing protein

→ Mia40이 SH → S-S 결합 유도 (disulfide bond)

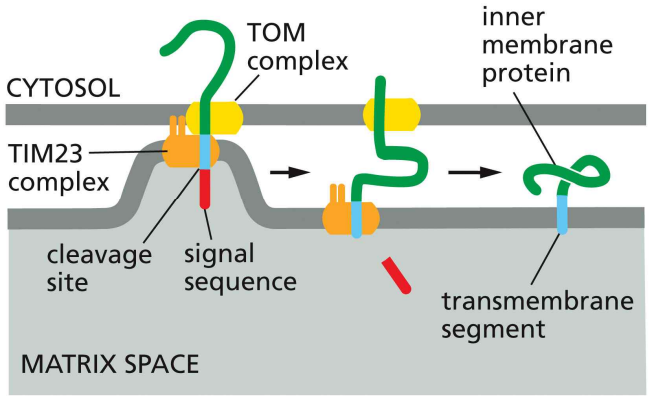
- 단백질이 부분적으로 folding → 안정화

→ **Back-sliding 방지 (trapping mechanism)**

- Respiratory chain이 Mia40을 reoxidation하여 반복 사용 가능

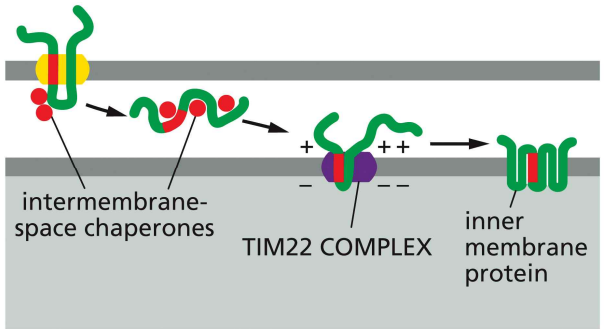
Figure 12–50 The role of energy in protein import into mitochondria.
Three different sources of energy are used to import protein into the mitochondria.

Transport Into the Inner Mitochondrial Membrane Occurs Via Several Routes



(A) The N-terminal signal sequence *(red)* initiates import into the matrix space

- N-terminal signal sequence (red) → matrix targeting
- 뒤에 오는 Hydrophobic transmembrane domain (blue, stop-transfer)
- Protein anchored in inner membrane (lateral insertion)

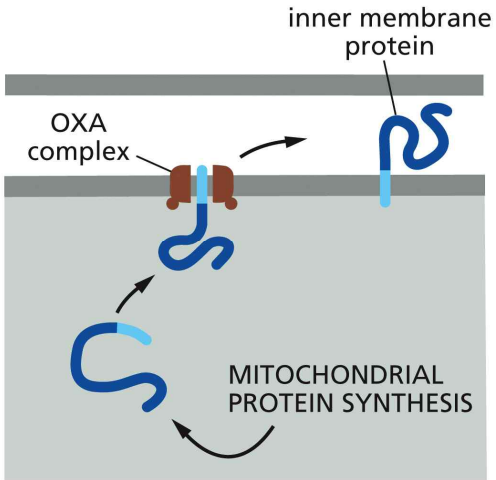


(B) Multipass inner membrane proteins that function as metabolite transporters.

- Internal signal sequences thread through TOM as loops
- Chaperones in intermembrane space guide proteins → TIM22 complex
- The TIM22 complex is specialized for the insertion of multipass inner membrane proteins.

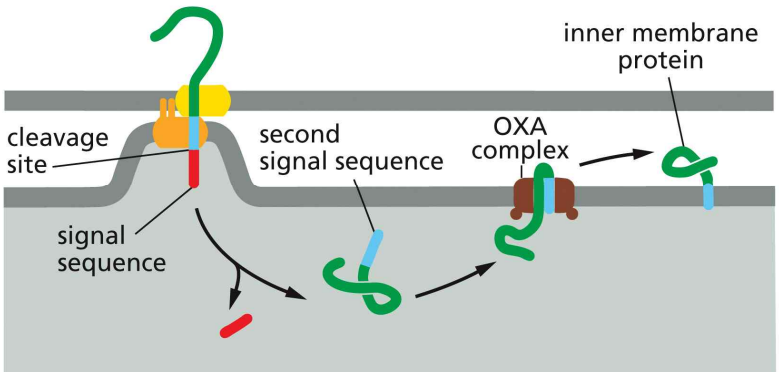
Figure 12–51 Routes for the production of inner mitochondrial membrane proteins.

Transport Into the Inner Mitochondrial Membrane Occurs Via Several Routes



(C) The OXA complex (Mitochondrial-encoded proteins)

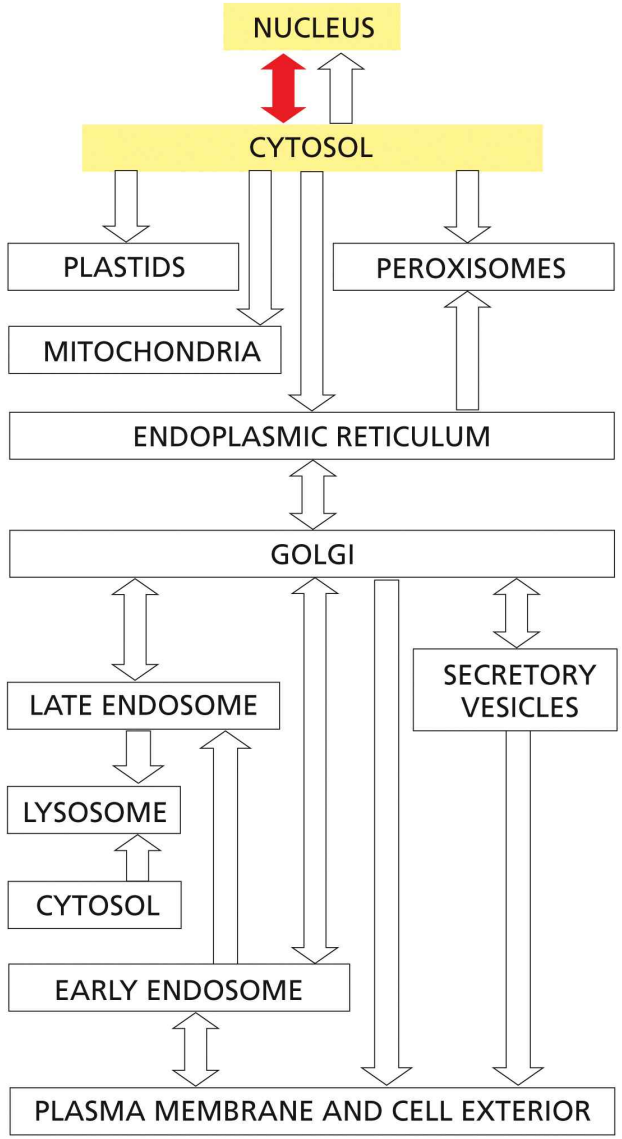
- Proteins translated in matrix from mitochondrial genome
- Inserted directly into inner membrane via **OXA complex**



(D) The OXA complex in the inner membrane can mediate protein insertion from the matrix space.

- Nuclear-encoded proteins imported fully into matrix
- Signal peptide cleaved → new hydrophobic signal exposed
- OXA complex inserts protein into inner membrane

Figure 12–51 Routes for the production of inner mitochondrial membrane proteins.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LYSOSOME
CYTOSOL
EARLY ENDOSOME
PLASMA MEMBRANE AND CELL EXTERIOR

Nuclear Pore Complexes Perforate the Nuclear Envelope

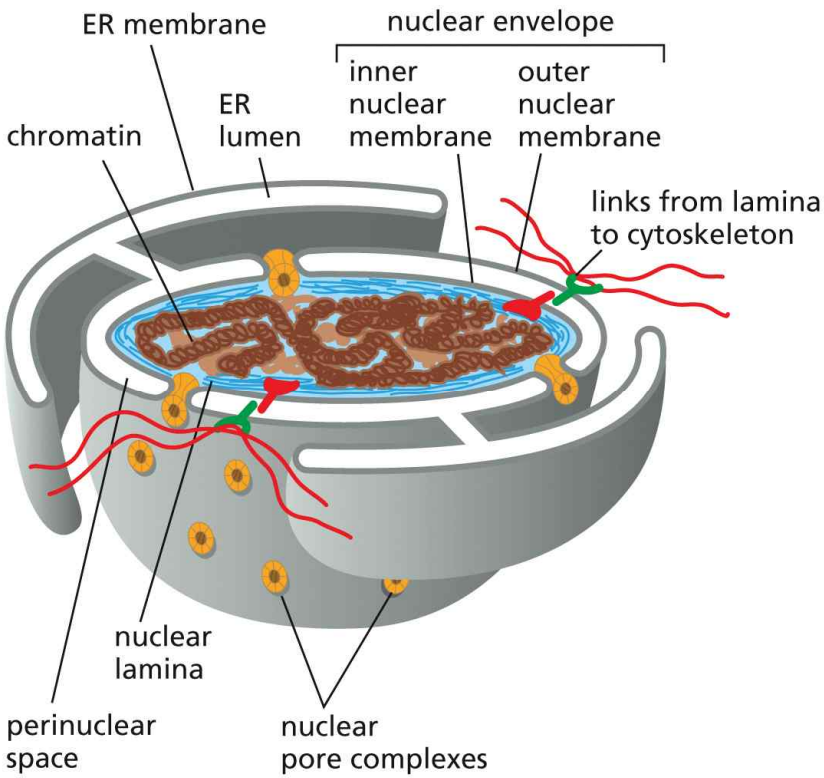

Figure 12–54A The nuclear envelope.

- ➢ 핵은 단순한 DNA 저장소가 아니라, 내부에 여러 **소기관 구조**를 가짐.
- ➢ 핵–세포질 간 **선택적 물질 이동**이 주요 기능

- **Nuclear Envelope (핵막)**
- Double membrane structure
  Outer membrane (외막) → Rough ER과 연속
  inner membrane (내막) → Nuclear lamina와 연결되어 핵 구조의 안정성 유지
- 두 막 사이 공간 → Perinuclear space
- 핵막에는 Nuclear pore complex(NPC) 존재 → 분자 이동 통로 역할

- **Nuclear pore complex (핵공복합체)**
- 매우 큰 multiprotein complex (~30 proteins, nucleoporins)
- 양방향 transport 가능 (signal-dependent transport)

- **Nucleolus (핵소체)**
- 리보솜 rRNA 합성과 리보솜 소단위체 조립이 일어나는 곳
- 단백질 수송과 직접적으로 연결: 리보솜 단위체는 핵에서 합성 → NPC를 통해 세포질로 수송

- **Nuclear lamins (핵라민)**
- 핵막 안쪽을 지지하는 intermediate filament protein (중간섬유 단백질)
- 핵의 구조 유지, chromatin organization, DNA 복제·전사 조절에도 관여

Nuclear Pore Complex

- **Transport properties**
- ~ 120 nm diameter large multiprotein complex
- 선택적 통로(selective gateway) 역할
  Small molecules → diffuse freely
  Macromolecules (proteins, RNA) → Signal-dependent transport

- **Symmetry**
- 8중 회전 대칭성 (eightfold rotational symmetry)
  NPC는 약 30여 종의 단백질(nucleoporin)로 구성
  반복된 대칭 구조(8, 16, 32 copies)를 통해 거대한 복합체를 형성

- 2중 횡단 대칭성 (twofold transverse symmetry)
  Nuclear vs cytosolic side → 양쪽 구조는 유사하지만 기능적으로는 다름

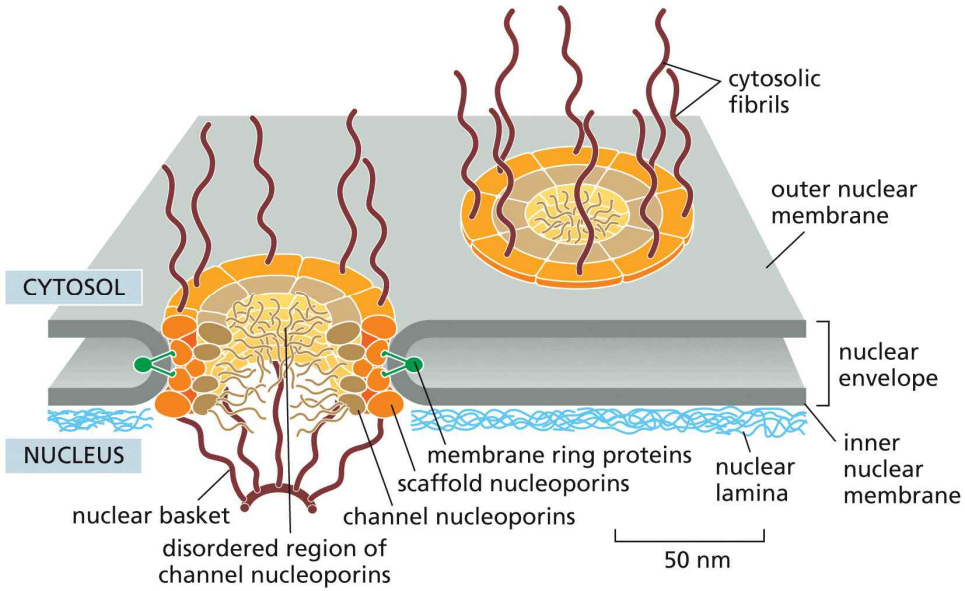

Figure 12–55A The arrangement of NPCs in the nuclear envelope

- **Nucleoporins classes**
- **Transmembrane ring proteins**: 핵막을 관통하며 NPC를 핵막에 고정
- **Scaffold nucleoporins**: Ring structure 형성, 일부는 막 굽힘 단백질(membrane-bending proteins) 역할 수행
→ NPC의 구조적 안정성 유지
- **Channel nucleoporins**: 중심 통로를 둘러싸며, 무질서한 불규칙 영역, Phenylalanine–glycine (FG) repeats (FG repeats) 이 풍부
→ Selective barrier (gel-like mesh) 을 형성하여 대형 분자의 비특이적 확산 차단

- **Peripheral structures**
- Fibrils protrude from both the cytosolic and the nuclear sides of the NPC
  Cytoplasmic filaments→ cargo recognition / docking
  Nuclear basket→ nuclear side에서 모여 nuclear basket 형성→ RNA processing / export 관련

➢ Size + signal에 의해 transport 결정

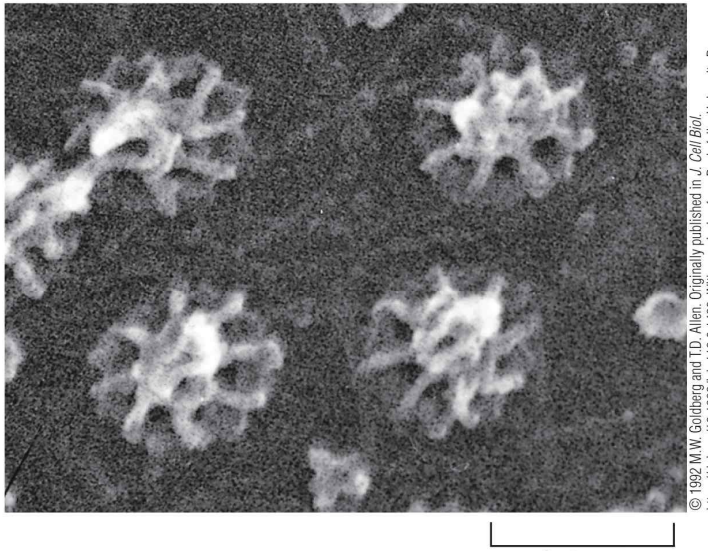

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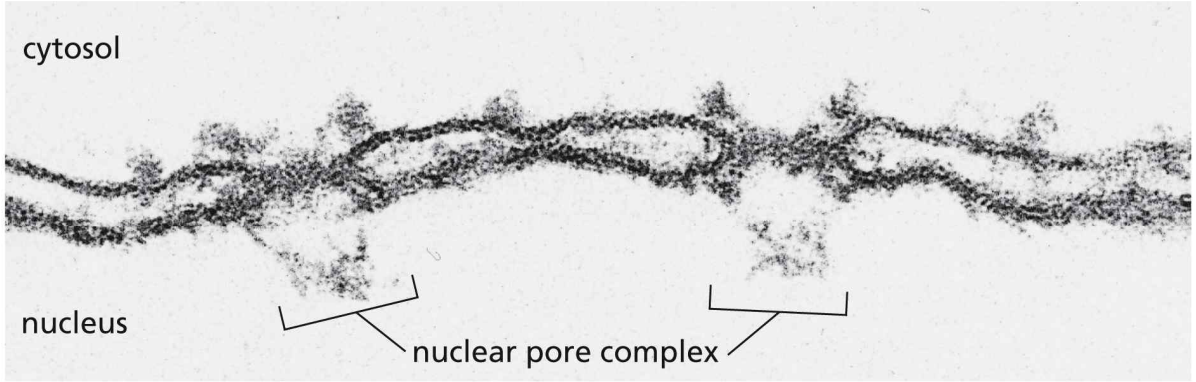
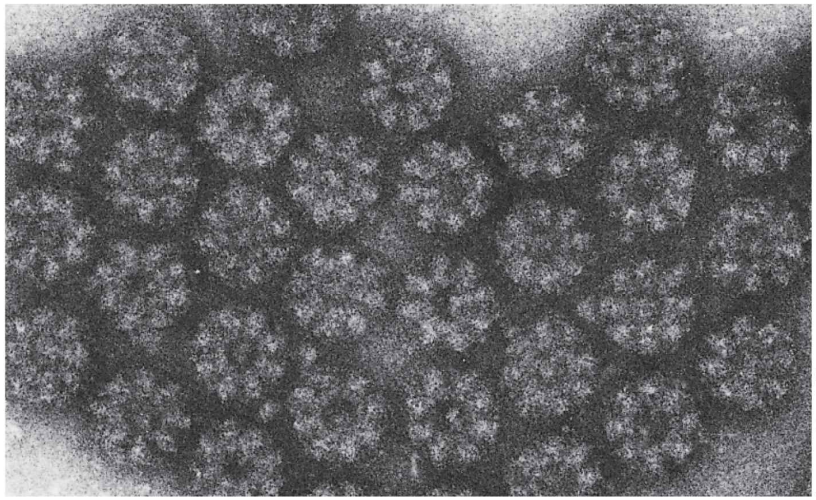
Figure 12–55 The arrangement of NPCs in the nuclear envelope.
(B) A scanning electron micrograph of the nuclear side of the nuclear envelope of an oocyte, showing NPCs with their basketlike fibrils.



Courtesy of Werner Franke and Ulrich Scheer

0.1 μm

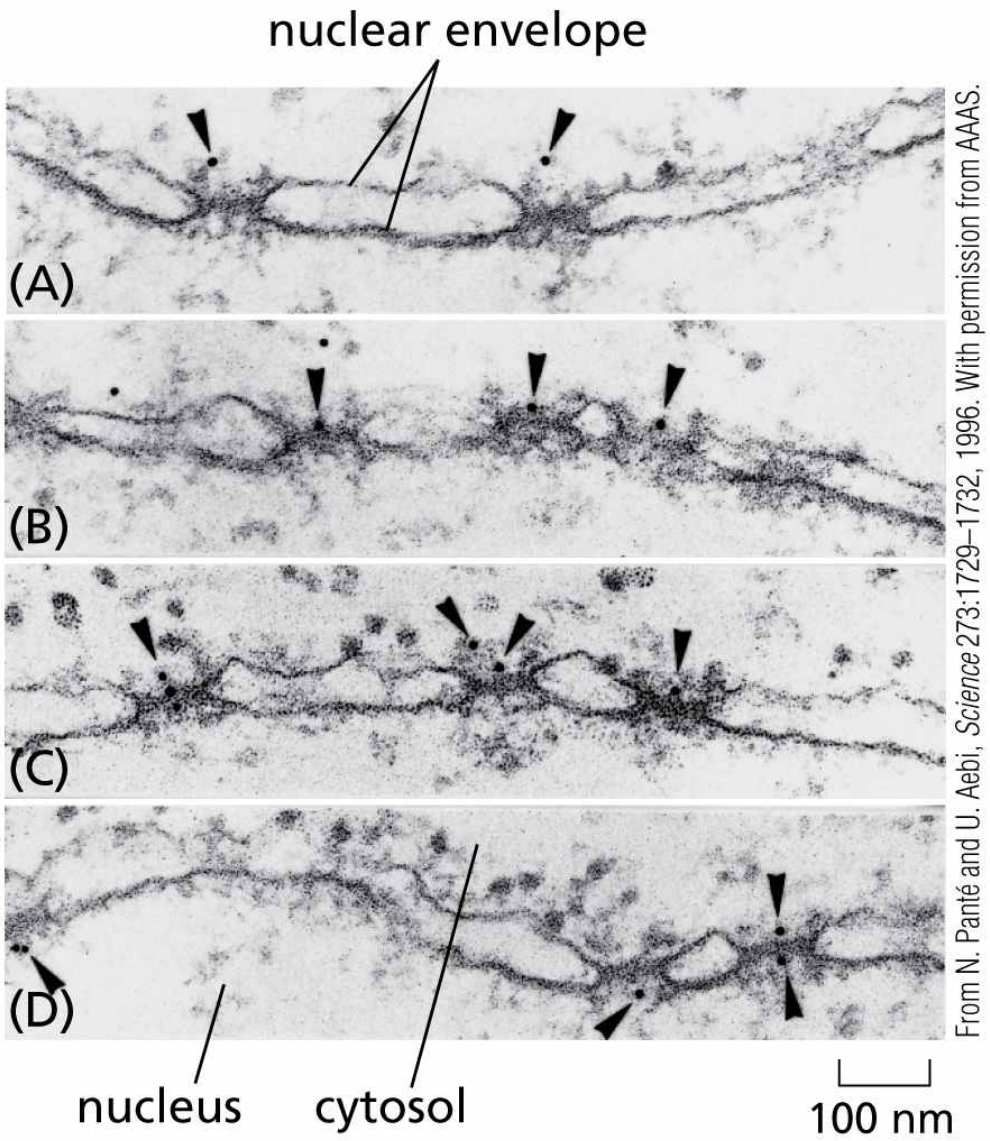
(**C**) An electron micrograph showing a side view of two NPCs *(brackets)*; note that the inner and outer nuclear membranes are continuous at the edges of the pore.



Courtesy of Ron Milligan

0.1 μm

(D) An electron micrograph showing face-on views of negatively stained NPCs. The membrane has been removed by detergent extraction. Note that some of the NPCs contain material in their center, which is thought to be trapped macromolecules in transit through these NPCs.



From N. Panté and U. Aebi, *Science* 273:1729–1732, 1996. With permission from AAAS.

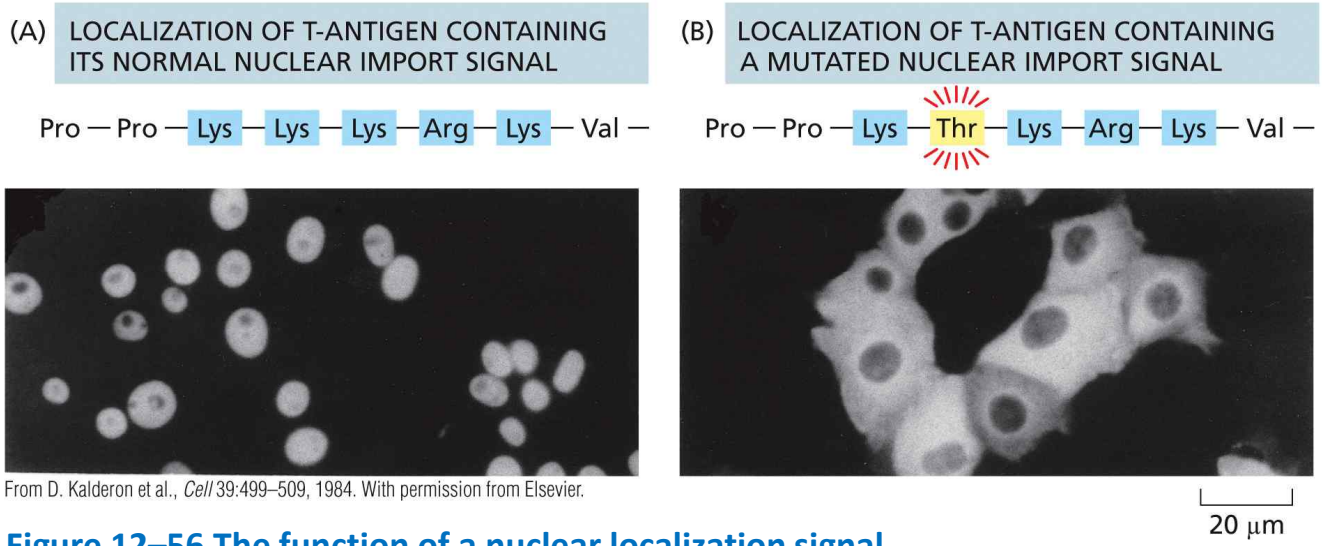
Figure 12–57 Visualizing active import through NPCs.
This series of electron micrographs shows 5- to 10-nm-diameter colloidal gold spheres *(arrowheads)* coated with peptides containing nuclear localization signals entering the nucleus through NPCs. The gold particles were injected into the cytosol of living cells, which then were fixed and prepared for electron microscopy at various times after injection. (**A**) Gold particles are first seen in proximity to the cytosolic fibrils of the NPCs. (**B, C**) They are then seen at the center of the NPCs, exclusively on the cytosolic face. (**D**) They then appear on the nuclear face. These gold particles have much larger diameters than those of the diffusion channels in the NPC and are imported by active transport.

Nuclear Pore Complex (NPC) and Molecular Transport

- Passive Diffusion (크기 제한 있음)
- < 5 kDa (~5 nm) → 자유 확산 가능 (이온, 대사산물 등)
- ~ 40 kDa (~9 nm) → 제한적/느린 확산
- 에너지 필요 없음, signal sequence 불필요

- Active Transport (수송 신호 필요)
- > ~60 kDa (~>9–10 nm) → Passive diffusion 불가
- NLS (Nuclear Localization Signal) 또는 NES (Nuclear Export Signal) 필요
- Importin / Exportin 필요
- 제한 없음 (large cargo 가능), signal-dependent + energy-dependent transport

Nuclear Localization Signals Direct Proteins to the Nucleus



From D. Kalderon et al., *Cell* 39:499–509, 1984. With permission from Elsevier.

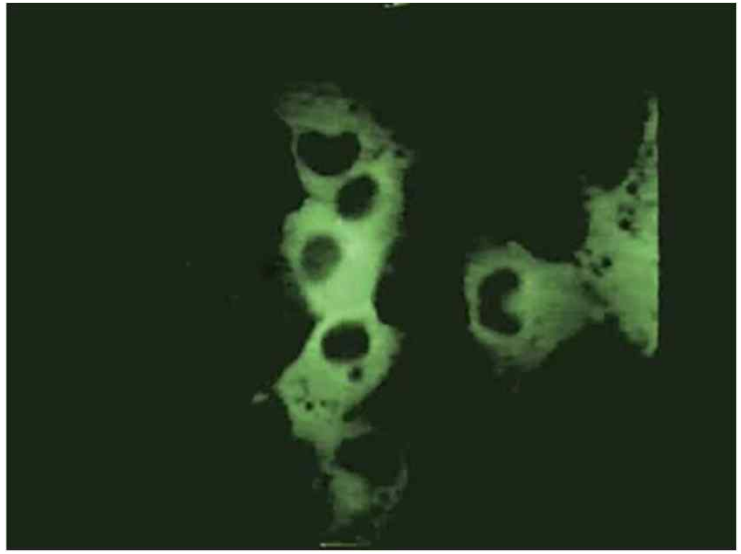


Figure 12–56 The function of a nuclear localization signal.
Immunofluorescence micrographs showing the cell location of SV40 virus T-antigen containing or lacking a short sequence that serves as a nuclear localization signal.

- **Nuclear Localization Signal (NLS)**
- **Lysine(K), arginine(R)**이 풍부한 basic sequence
- Importin(α/β complex)이 NLS를 인식 → 단백질이 NPC를 통해 핵 안으로 import

- **Nuclear Export Signal (NES)**
- 핵에서 세포질로 나가야 할 단백질/ RNA binding protein에 있는 **Leucine-rich sequence**
- Exportin이 인식하여 단백질/RNA 세포질로 NPC를 통해 수출(export).

➢ 정밀한 제어: 수송 여부는 Ran GTPase cycle, phosphorylation, 단백질 결합 등에 의해 조절
➢ 전사인자(Transcription factors)는 필요할 때만 핵 안으로 들어가서 DNA와 작용
➢ RNA 및 리보솜 단백질은 합성 후 NES를 통해 세포질로 이동

Nuclear Import Receptors Bind to Both Nuclear Localization Signals and NPC Proteins

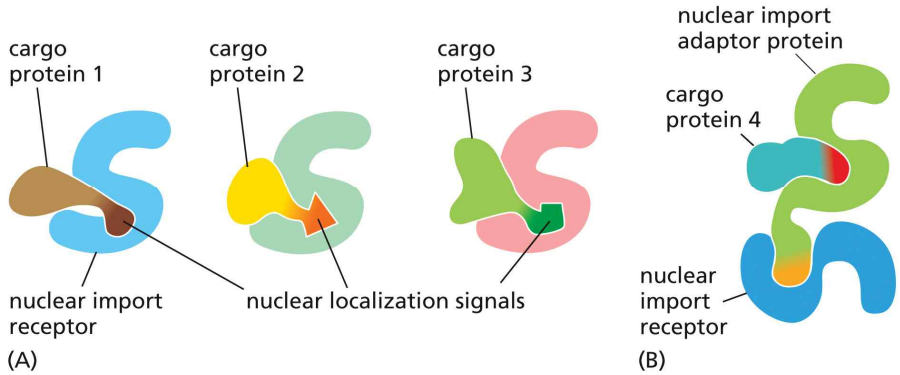


Figure 12–58 Nuclear import receptors.

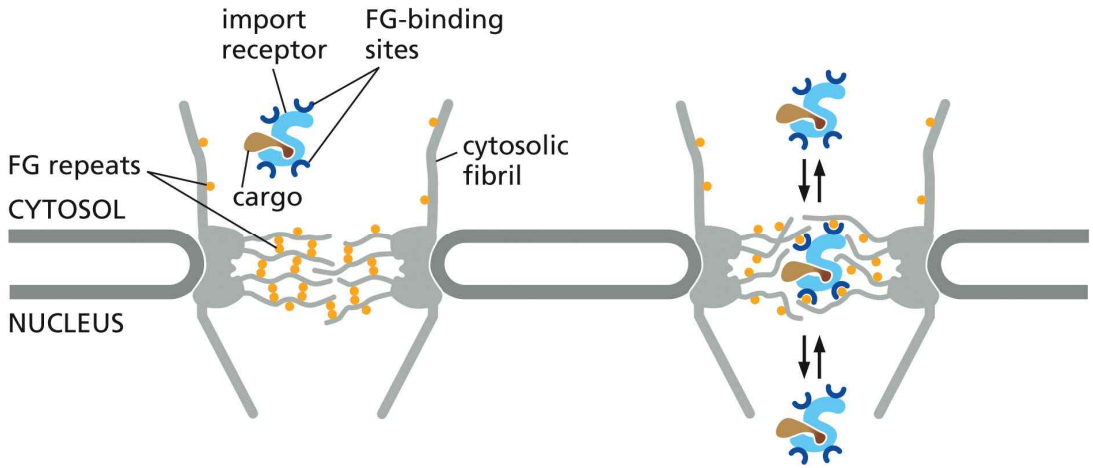


Figure 12–59 Interaction of nuclear import receptors with FG repeats.

Nuclear Import Receptors (Importins)

- 서로 다른 **NLS (Nuclear Localization Signal)** → 특정 import receptor가 인식
- Importin α (adaptor) → NLS 직접 인식
- Importin β (receptor) → NPC와 상호작용하여 transport
- 일부 단백질(cargo)은 바로 receptor와 결합할 수 없고, adaptor 단백질을 필요
- Adaptor 단백질은 receptor와 유사한 구조를 가지며, NLS를 인식해 receptor에 전달하는 역할

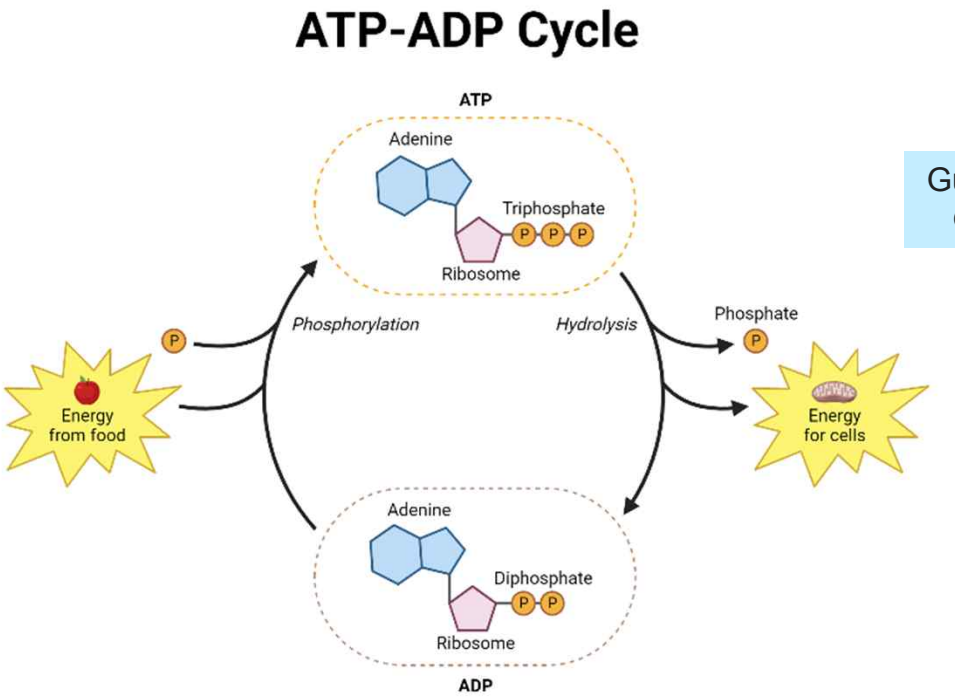
FG Repeats와의 상호작용

- NPC 내부는 FG repeat (Phe-Gly 반복) 단백질로 구성
- FG repeats create a selective barrier: Nonspecific protein diffusion 차단 (일반 단백질은 통과하지 못함)
- Import receptor는 FG repeat에 낮은 친화력으로 반복적으로 결합/해리하면서 mesh를 통과
- 이렇게 receptor에 결합된 cargo 단백질은 빠르고 효율적으로 핵 안으로 이동

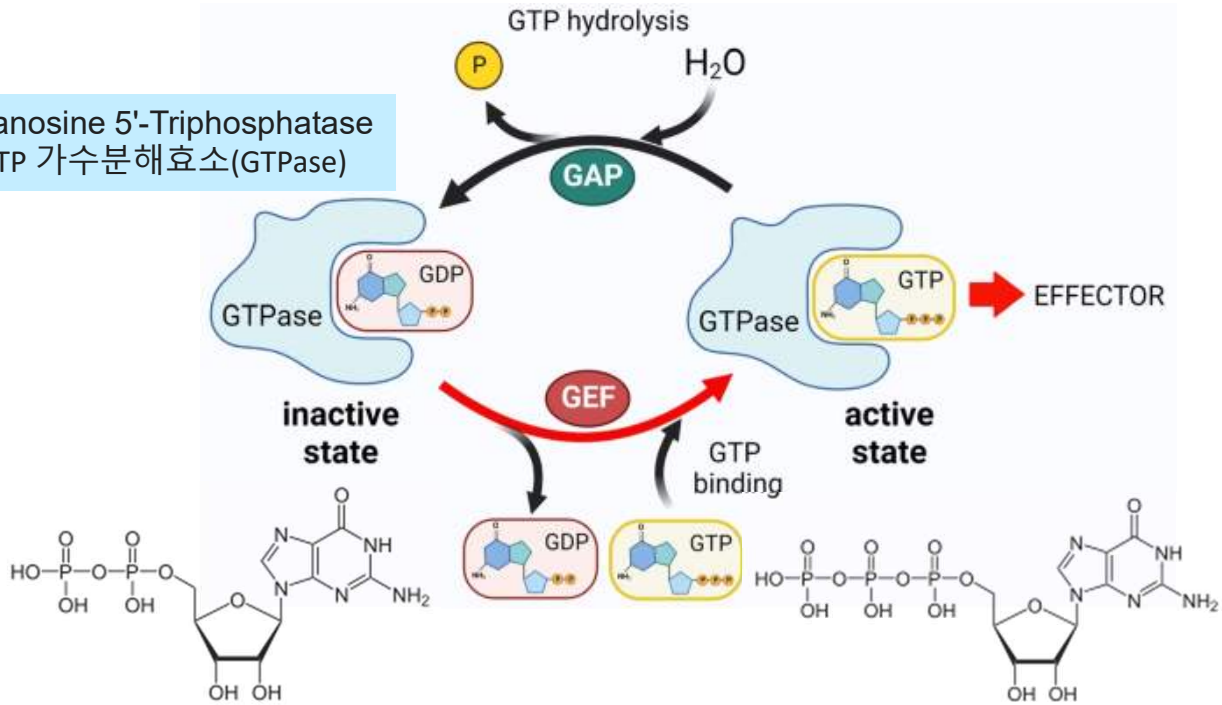
➢ Large proteins require **NLS + receptor-mediated active transport**.

➢ NPCs are not open holes, but **selective gates controlled by FG-repeat proteins**.

GTP-GDP cycle



Guanosine 5'-Triphosphatase
GTP 가수분해효소(GTPase)



Guanosine 5’-diphosphate

Guanosine 5'-Triphosphate

The Ran GTPase Imposes Directionality on Nuclear Import Through NPCs

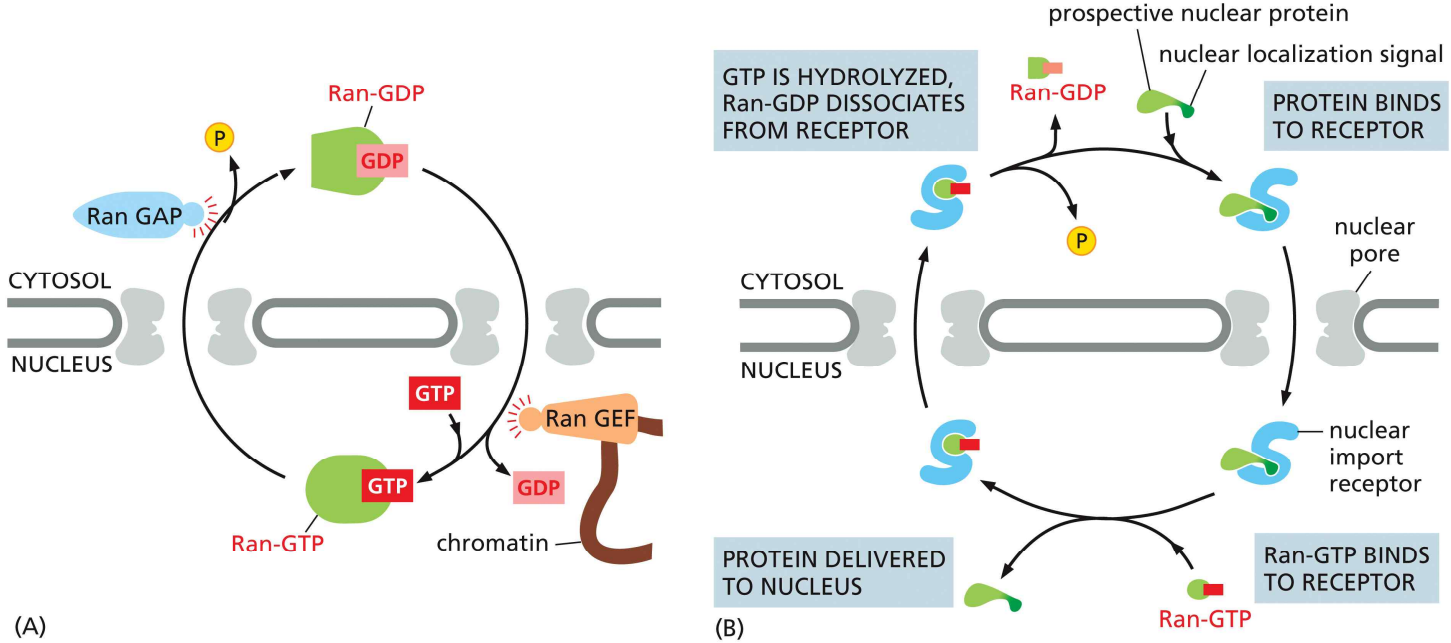


Figure 12–60 The compartmentalization of Ran-GDP and Ran-GTP provides directionality to nuclear transport.

- **Ran (**Guanosine 5'-Triphosphatase; **GTPase)**
- **Ran-GDP** (inactive): 주로 세포질(cytosol) 에 존재
- **Ran-GTP** (active): 주로 핵(nucleus) 에 존재
- ✓ RanGAP과 RanGEF의 공간적 분리가 Ran-GTP/Ran-GDP 구배를 형성하고 유지 → 방향성(directionality)

- **RanGAP (Ran GTPase-activating protein)**: Cytosol
- Ran-GTP → Ran-GDP로 가수분해
- **RanGEF (Ran guanine nucleotide exchange factor)**: Nucleus, chromatin-bound
- Ran-GDP → Ran-GTP 교환반응

Nuclear Export Works Like Nuclear Import, but in Reverse

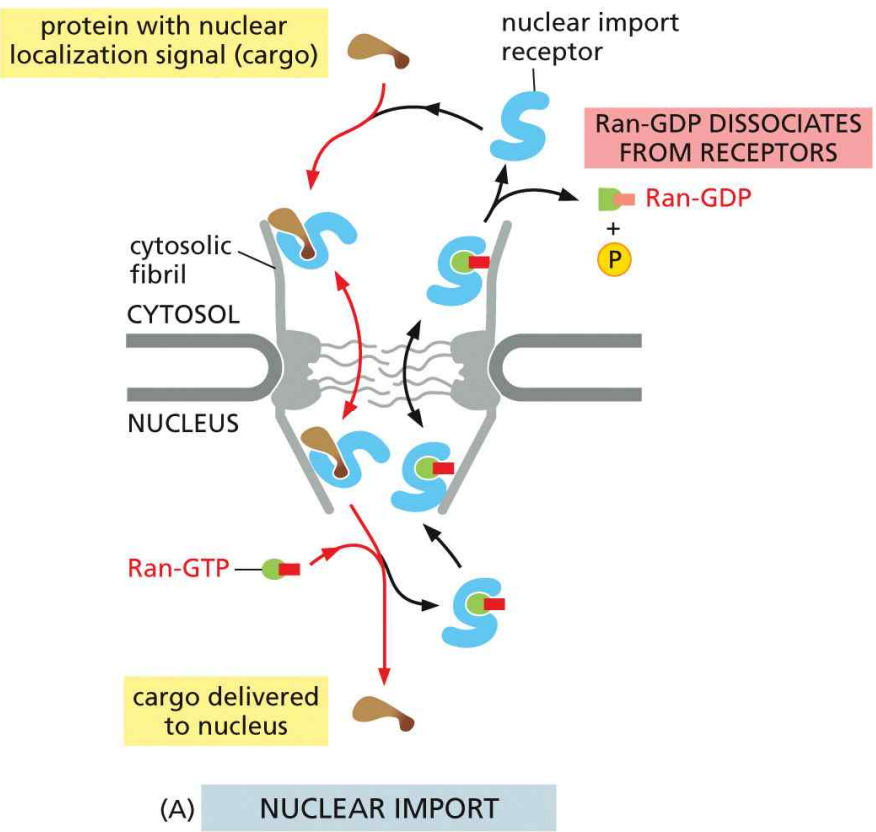


Figure 12–61 Nuclear import and nuclear export both use the Ran GTPase cycle.

Nuclear Import

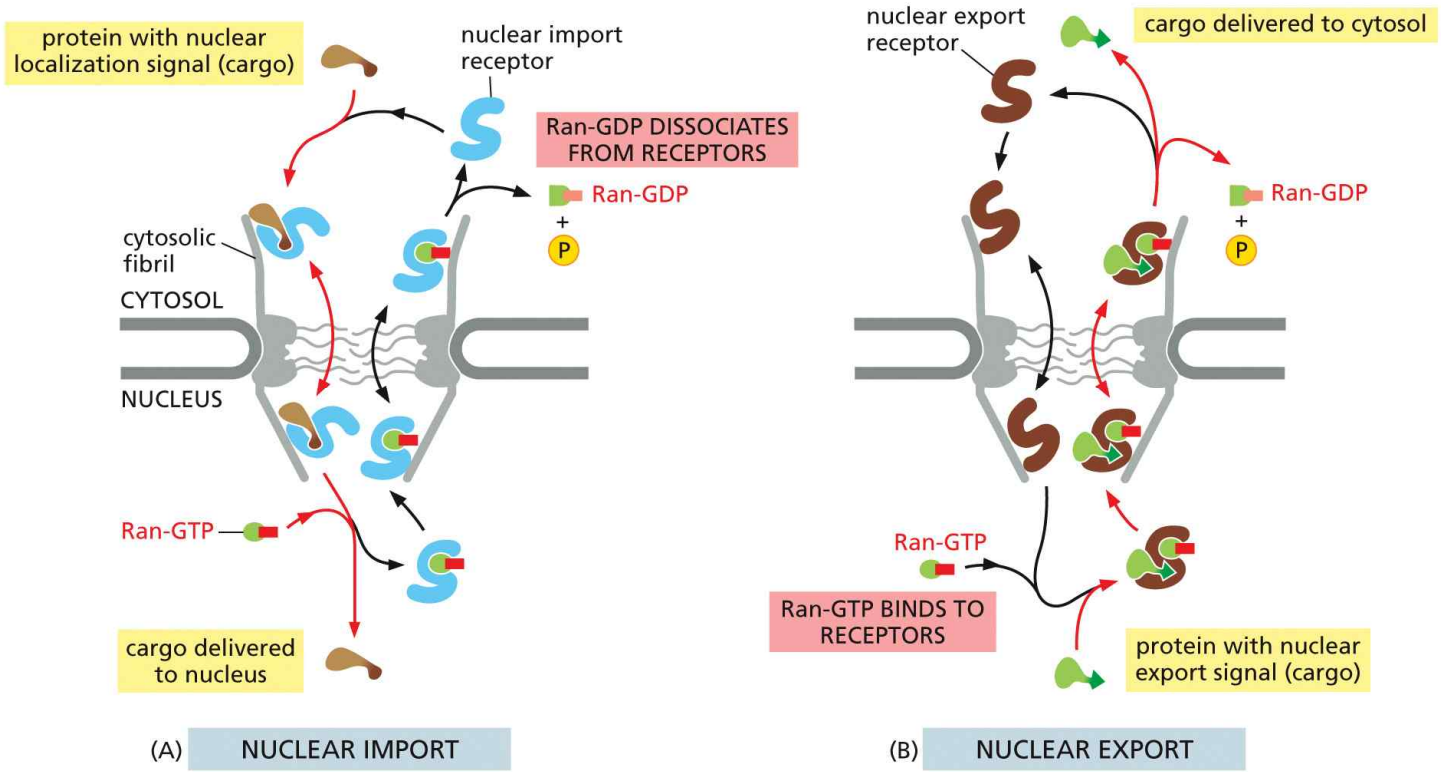
- **Cytosol**
  - Cargo protein이 nuclear localization signal (NLS)를 가짐
  - Importin (nuclear import receptor) 이 cargo를 인식
  - Cargo-importin complex가 nuclear pore complex (NPC)를 통과

- **Nucleus**
  - Ran-GTP가 importin에 결합 → cargo 방출
  - Cargo remains in nucleus

- **Cytosol**
  - Importin–Ran-GTP complex가 cytosol로 이동
  - RanGAP 작용: Ran-GTP → Ran-GDP
  - RanGAP이 Ran-GTP → Ran-GDP로 전환

  → Importin 해방 및 재활용

➢ 세포질에서는 수용체–cargo 결합이 선호됨
➢ 핵에서는 Ran-GTP 때문에 결합이 불리(disfavored) → cargo 방출
➢ 이 결합 친화도의 차이가 바로 한쪽 방향(세포질 → 핵)으로만 cargo가 들어가도록 만드는 힘



과정	Ran-GTP 역할
Import	Cargo release
Export	Complex formation

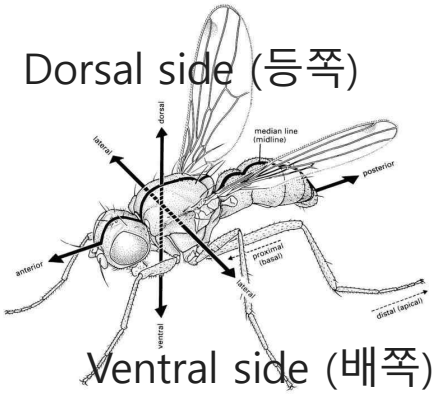
Nuclear export

- 핵에서 세포질로 protein이나 RNA 복합체는 내보낼 때도 NPC와 Ran 시스템이 이용
- Export receptor (exportin) 이 nuclear export signal (NES)를 가진 cargo를 인식

- **Nucleus**
- Ran은 주로 Ran-GTP 상태→ Exportin –cargo–Ran-GTP complex (trimeric complex)를 형성
- NPC를 통해 cytosol로 이동

- **Cytosol**
- Cytosol에서는 RanGAP이 Ran-GTP를 Ran-GDP로 가수분해
- RanGTP → Ran GDP 전환과 동시에 export complex가 붕괴되어 cargo가 cytosol에 방출

Transport Through NPCs Can Be Regulated by Controlling Access to the Transport Machinery



- 초기 초파리 배아는 다핵 공동세포질 (Syncytium stage) 상태
- Dorsal protein production: Dorsal 단백질은 cytoplasm 전체에 균일하게 합성되지만, 기능을 하려면 nucleus 안으로 들어가야 함
- 차별적인 핵 내 수송 조절 덕분에 배아의 등–배 축(dorsal–ventral axis)

→ 핵수송의 조절이 배아 발달의 방향성과 패턴 형성을 결정

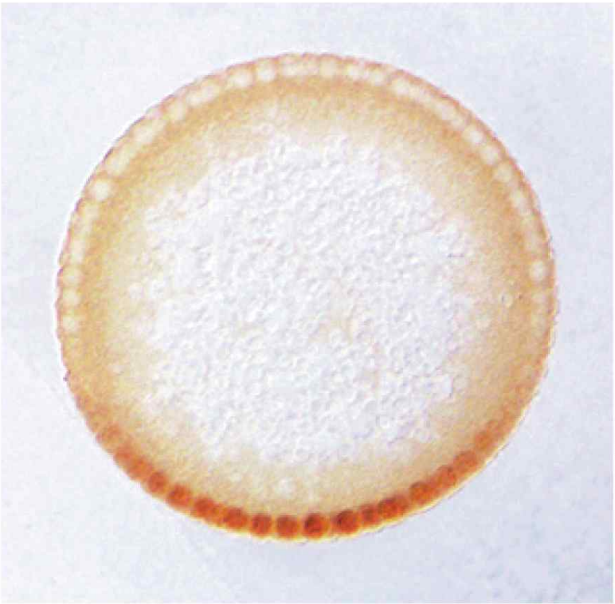


Figure 12–62 The control of nuclear transport in the early *Drosophila* embryo. The embryo at this stage is a syncytium, shown here in cross section, with many nuclei in a common cytoplasm, arranged around the periphery, just beneath the plasma membrane. The transcription regulatory protein Dorsal is produced uniformly throughout the peripheral cytoplasm, but it can act only when inside the nuclei. The Dorsal protein has been stained with an enzyme-coupled antibody that yields a brown product, revealing that Dorsal is excluded from the nuclei at the dorsal side *(top)* of the embryo but is concentrated in the nuclei toward the ventral side *(bottom)* of the embryo. The regulated traffic of Dorsal into the nuclei controls the differential development between the back and belly of the animal. (Courtesy of Siegfried Roth.)

NF-AT Nuclear Transport in T Cell Activation

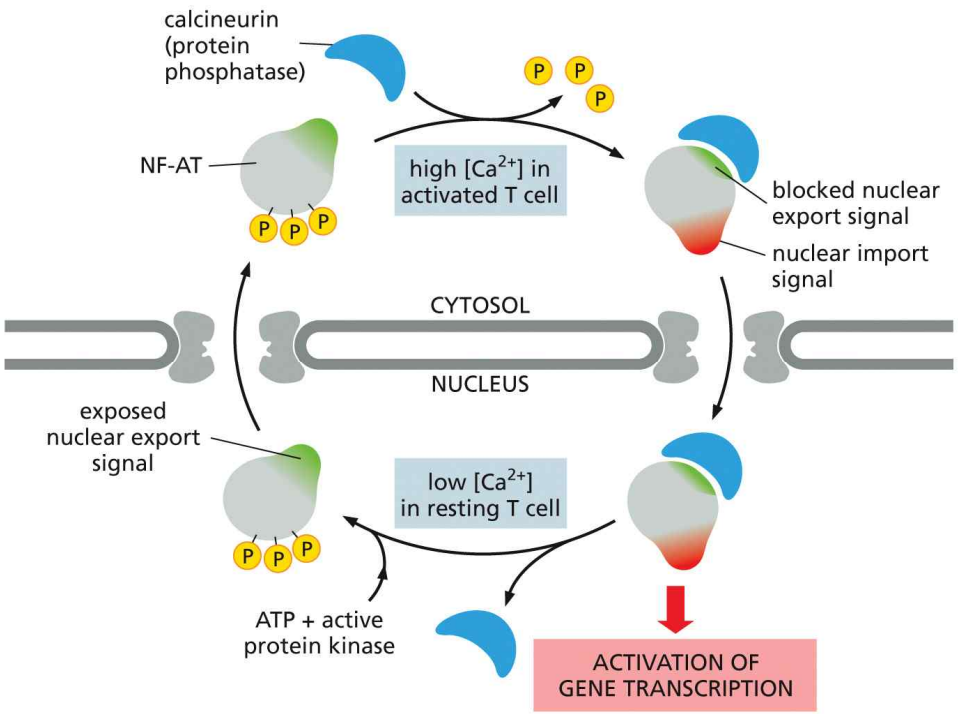


Figure 12–63 The control of nuclear import during T cell activation.

- **Resting T cell (비활성 상태)**
- NF-AT (전사 조절 단백질): 세포질에 인산화된 상태로 존재
- NES is exposed (active)/NLS is masked (inactive)

- **T cell activation (Ca^{2+} 증가)** :
- Antigen stimulation 항원 자극 → 세포 내 Ca^{2+} 농도 증가
- Calcineurin (Ca^{2+} -dependent phosphatase) 활성화

→ NF-AT dephosphorylation (탈인산화)

- **탈인산화 (dephosphorylation) 효과**
- NLS is exposed (active)/NES is masked (inactive)
- NF-AT–Calcineurin 복합체가 핵으로 이동
- Gene expression for T cell activation

- **Ca^{2+} 감소 시 (off switch)**
- Calcineurin이 NF-AT에서 분리
- NES is exposed (active)/NLS is masked (inactive)
- NF-AT rephosphorylation (재인산화)
- NF-AT이 세포질로 다시 이동

➢ NF-AT의 세포 내 위치는 인산화 상태에 따른 NLS/NES의 노출 변화와 Ca^{2+} 신호에 의해 결정
➢ T cell의 면역반응은 정밀하게 제어
➢ CyclosporinA, FK506 (Tacrolimus) → calcinurin 억제
NF-AT dephosphorylation 억제 → NF-AT의 핵 이동 차단→ T cell activation 억제

Figure 12–63 The control of nuclear import during T cell activation. The nuclear factor of activated T cells (NF-AT) is a transcription regulatory protein that, in the resting T cell, is found in the cytosol in a phosphorylated state. When T cells are activated by foreign antigen (discussed in Chapter 24), the intracellular Ca^{2+} concentration increases. In high Ca^{2+}, the protein phosphatase calcineurin binds to NF-AT and dephosphorylates it. The dephosphorylation exposes nuclear import signals and blocks a nuclear export signal. The complex of NF-AT and calcineurin is therefore imported into the nucleus, where NF-AT activates the transcription of numerous genes required for T cell activation. The response shuts off when Ca^{2+} levels decrease, releasing NF-AT from calcineurin. Rephosphorylation of NF-AT inactivates the nuclear import signals and reexposes the nuclear export signal, causing NF-AT to relocate to the cytosol. Some of the most potent immunosuppressive drugs, including cyclosporin A and FK506, inhibit the ability of calcineurin to dephosphorylate NF-AT and thereby block the nuclear accumulation of NF-AT and T cell activation (Movie 12.1).

The Nuclear Envelope Disassembles and Reassembles During Mitosis

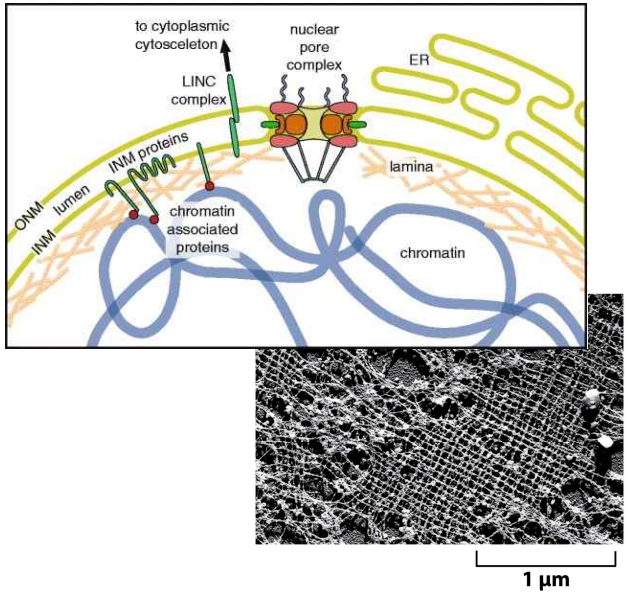
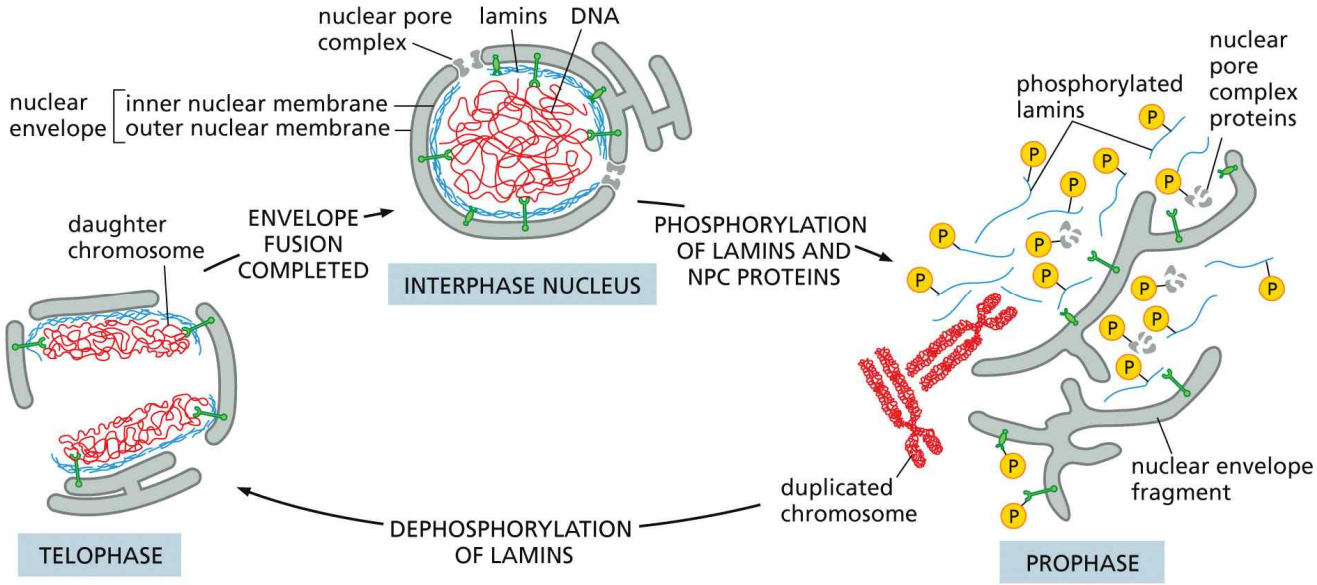


Figure 12–65 The breakdown and re-formation of the nuclear envelope and lamina during mitosis.



- **Nuclear Envelope Breakdown**
- CDK1–cyclin B kinase activation (M phase)
- 라민(lamin) 단백질의 인산화(phosphorylation) → Nuclear lamina가 해체 → Nuclear envelope breakdown (핵막 붕괴)
- 일부 nucleoporin과 inner nuclear membrane proteins도 같은 방식으로 인산화되어 붕괴 과정에 기여

- **Nuclear Envelope Re-formation**
- 세포분열 후기(anaphase/telophase)에 라민 dephosphorylation → lamina 재조립 시작
- Nuclear lamina가 분리된 염색체(decondensing daughter chromosomes) 주변에서 다시 형성됨.
- ER-derived membrane vesicles → chromatin에 결합
- Lamin receptors를 가진 막 조각이 라민 네트워크에 결합.
- Membrane fusion을 통해 하나의 완전한 nuclear envelope 재형성

3월 6일(금) ~ 6월 12일(금)					
주차	날짜	수업명	교수자	소속	비고
1	3/6	Introduction Visualizingcell	박용환	미생물학교실	강의
2	3/13	Membranestructure	박용환	미생물학교실	강의
3	3/20	Intracellularcompartmentsandproteinsorting	모정순	의과학연구소	강의
4	3/27	Intracellularcompartmentsandproteinsorting	모정순	의과학연구소	강의
5	4/3	IntracellularMembraneTraffic	박용환	미생물학교실	강의
6	4/10	IntracellularMembraneTraffic	박용환	미생물학교실	강의
7	4/17	Cytoskeleton	장재락	뇌과학교실	강의
8	4/24	Cytoskeleton	장재락	뇌과학교실	강의
9	5/1	중간고사 (여유당)	모정순	의과학연구소	
10	5/8	Mitochondria	박찬배	생리학교실	강의
11	5/15	Mitochondria	박찬배	생리학교실	강의
12	5/22	Mitochondria	박찬배	생리학교실	강의
13	5/29	CellJunction&ECM	이은정	뇌과학교실	강의
14	6/5	CellJunction&ECM	이은정	뇌과학교실	강의
15	6/12	기말고사 (여유당)	모정순	의과학연구소	

형성평가 질문지 작성
(아주 Bb)